Instructions: Show all work in the space provided. Write each final answer in its box. Only what is inside a box is graded. Write the letter of the single best answer in its Letter box. Write intervals in interval notation. No notes. **70 points total.**

Part A — Rates of Change and Characteristics (14 pts)

Problems 1 and 2 use this table of values for a polynomial f.

x	0	1	2	3	4
$f(x)$	1	-2	1	16	49

1. Use successive differences to find the degree of the polynomial that fits f.

1st differences : $-3,\ 3,\ 15,\ 33$
2nd differences : $6,\ 12,\ 18$
3rd differences : $6,\ 6$

Degree: 3

2. Find the average rate of change of f on the interval $[1, 4]$.

$$\frac{f(4) - f(1)}{4 - 1} = \frac{49 - (-2)}{3} = 17$$

17

3. Let $g(x) = x^2 - 3x$. Find and simplify the difference quotient $\dfrac{g(x+h) - g(x)}{h}$, $h \neq 0$.

$$g(x+h) = (x+h)^2 - 3(x+h)$$
$$= x^2 + 2xh + h^2 - 3x - 3h$$
$$g(x+h) - g(x) = 2xh + h^2 - 3h$$
$$\frac{g(x+h) - g(x)}{h} = 2x + h - 3$$

$2x + h - 3$

4. The graph of the polynomial f continues in the direction of its arrows. Its turning points are marked.

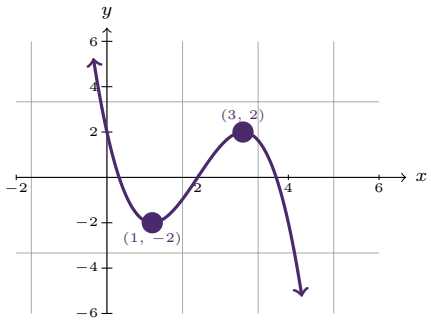
(a) The y-intercept: (0, 2)

(b) The relative maximum value: 2

(c) Increasing on: $(1, 3)$

(d) Decreasing on: $(-\infty, 1) \cup (3, \infty)$

(e) The point of inflection: (2, 0)



(f) Which is true?
(A) The absolute maximum of f is 2.
(B) The absolute maximum of f is -2.
(C) f has no absolute maximum.

Letter: C

Part B — Polynomial Behavior (12 pts)

5. Let $f(x) = 3x^2 - 2x^5 + x - 4$.

Degree	Leading coefficient	Most turning points	Most real zeros
5	-2	4	5

6. Find the leading term of $g(x) = (1 - 2x)(x + 3)^3$.

$(-2x)(x)^3 = -2x^4$

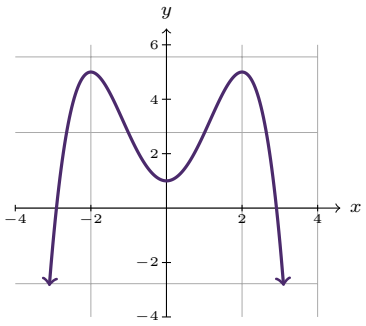
Leading term: $-2x^4$

7. Give the end behavior of the same g in limit notation.

$\lim_{x \to -\infty} g(x) =$ $-\infty$

$\lim_{x \to \infty} g(x) =$ $-\infty$

8. The graph of the polynomial p is shown. It continues in the direction of its arrows.



(a) The least possible degree of p: 4

(b) Which describes the degree and the leading coefficient of p?
(A) even, positive (B) even, negative
(C) odd, positive (D) odd, negative

Letter: **B**

Part C — Zeros and Multiplicity (10 pts)

9. Let $h(x) = (x + 3)^2(x - 1)(x - 4)^3$. Complete the table. Write *crosses* or *touches* for what the graph does at each zero.

Zero of h	-3	1	4
Multiplicity	2	1	3
Crosses or touches?	**touches**	**crosses**	**crosses**

10. For the same h, is $h(x)$ *positive* or *negative* just to the left of $x = -3$? Just to the right of $x = -3$?

$x \approx -3 \Rightarrow (x - 1) < 0,\ (x - 4)^3 < 0$
$h(x) > 0$ **on both sides**

Just left: **positive** Just right: **positive**

11. A polynomial of degree 3 with real coefficients has zeros 4 and $1 + 2i$. What is its third zero?

$1 - 2i$

Part D — Factoring and Solving (11 pts)

For 12–14, factor completely.

12. $2x^3 - 18x$

$$2x^3 - 18x = 2x(x^2 - 9)$$
$$= 2x(x - 3)(x + 3)$$

$2x(x-3)(x+3)$

13. $x^3 + 3x^2 - 4x - 12$

$$x^3 + 3x^2 - 4x - 12 = x^2(x + 3) - 4(x + 3)$$
$$= (x + 3)(x^2 - 4)$$
$$= (x + 3)(x - 2)(x + 2)$$

$(x+3)(x-2)(x+2)$

14. $x^3 + 64$

$$x^3 + 64 = x^3 + 4^3$$
$$= (x + 4)(x^2 - 4x + 16)$$

$(x+4)(x^2-4x+16)$

15. Solve $x^4 - 5x^2 - 36 = 0$. Give every solution, real and non-real.

$$(x^2 - 9)(x^2 + 4) = 0$$
$$x^2 = 9 \text{ or } x^2 = -4$$
$$x = \pm 3 \text{ or } x = \pm 2i$$

$x =$ $3,\ -3,\ 2i,\ -2i$

16. To solve $x^3 = 16x$, Dana divides both sides by x and gets $x^2 = 16$, so $x = \pm 4$. Which statement is true?

(A) Dana is right: the solutions are 4 and -4.
(B) Dana lost a solution: the solutions are 0, 4, and -4.
(C) Dana lost a solution: the solutions are 16, 4, and -4.
(D) Dana gained a solution: the only solution is 4.

Letter: **B**

Part E — Dividing Polynomials (14 pts)

17. Use long division to divide $2x^3 + x^2 - 4x + 3$ by $x^2 + 1$. Fill in the boxes.

$$2x^3 + x^2 - 4x + 3 - 2x(x^2 + 1) = x^2 - 6x + 3$$
$$x^2 - 6x + 3 - 1(x^2 + 1) = -6x + 2$$

$2x^3 + x^2 - 4x + 3 = (x^2 + 1)($ $\boxed{2x + 1}$ $) + \boxed{-6x + 2}$

18. Use synthetic division to divide $2x^3 - 5x + 7$ by $x + 2$.

-2	2	0	-5	7
		-4	8	-6
	2	-4	3	1

Quotient: $\boxed{2x^2 - 4x + 3}$

Remainder: $\boxed{1}$

19. Use the Remainder Theorem to find the remainder when $p(x) = x^4 - 3x^3 + 2x - 5$ is divided by $x - 2$.

$$p(2) = 16 - 24 + 4 - 5$$
$$= -9$$

Remainder: $\boxed{-9}$

20. $x = 2$ is a zero of $p(x) = x^3 - 3x^2 + 4$. In the box: divide $p(x)$ by $x - 2$, write $p(x)$ factored completely, and list every zero of p with its multiplicity.

Synthetic division by 2 on $1,\ -3,\ 0,\ 4$ gives $1,\ -1,\ -2$ with remainder 0, so $x - 2$ is a factor. Quotient $x^2 - x - 2 = (x - 2)(x + 1)$. So $p(x) = (x - 2)^2(x + 1)$. Zeros: 2 (multiplicity 2) and -1 (multiplicity 1).

Part F — Writing Polynomials from Zeros (9 pts)

21. Write the polynomial with zeros -1, 2, and 3 and leading coefficient 1, in standard form.

$$\begin{aligned}(x+1)(x-2)(x-3) &= (x+1)(x^2-5x+6)\\ &= x^3-5x^2+6x+x^2-5x+6\\ &= x^3-4x^2+x+6\end{aligned}$$

x^3-4x^2+x+6

22. A polynomial f of degree 3 has zero 2 with multiplicity 2, zero -1, and passes through $(0,8)$. Find the leading coefficient a, then write $f(x)$ in factored form.

$$\begin{aligned}f(x) &= a(x-2)^2(x+1)\\ f(0) = a(-2)^2(1) &= 8\\ 4a &= 8\\ a &= 2\end{aligned}$$

$a =$ 2 $\qquad f(x) =$ $2(x-2)^2(x+1)$

23. Write the polynomial of least degree with real coefficients, leading coefficient 1, and zeros 3 and $2i$, in standard form.

$$\begin{aligned}2i \text{ is a zero} &\Rightarrow -2i \text{ is a zero}\\ (x-3)(x-2i)(x+2i) &= (x-3)(x^2+4)\\ &= x^3-3x^2+4x-12\end{aligned}$$

$x^3-3x^2+4x-12$